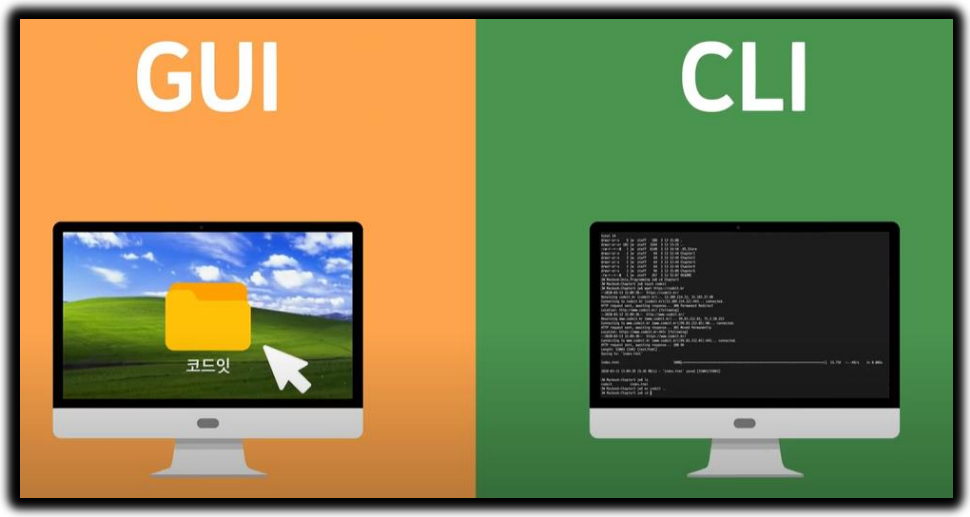
WINDOWS COMMANDLINE PART 1

The Command Line's Power:

The command line remains a **powerful tool for system administrators** and users, despite the GUI taking center stage. While the GUI can be convenient, it doesn't give access to everything you need, and relying on it for certain tasks can be inefficient and error-prone.



Automation and Efficiency:

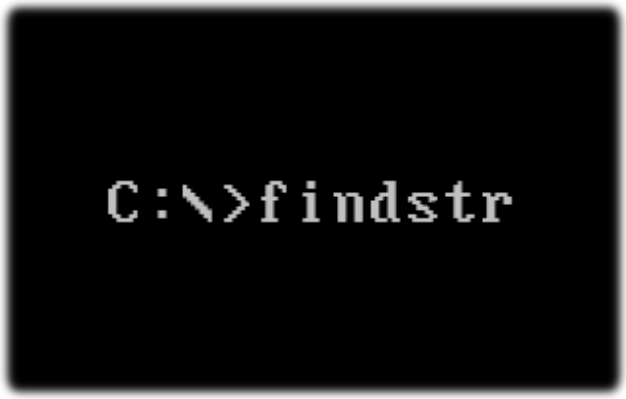
The command line allows for **automation of tasks**, saving time and reducing the chances of mistakes. For instance, once you learn the right commands, you can execute multiple tasks quickly without repeating actions manually as you would in the GUI.

Windows PowerShell and the Command Line:

Though **PowerShell is increasingly being used**, the command line is still very much supported in Windows Vista. In fact, Vista introduced significant updates to the command line, and the book highlights these changes.

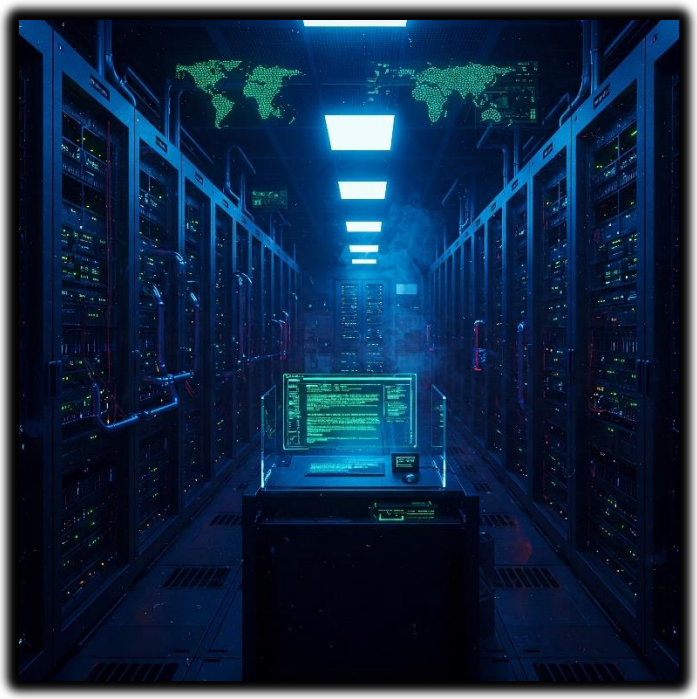
Why the Command Line is Better:

The **GUI often fails in searching for specific files or tasks**, especially when it comes to searching executables. Tools like the **FindStr** utility (explained in Chapter 4) offer high-speed searching that the GUI can't match.



Detailed Information and Real-World Use:

The book doesn't just teach basic command usage; it provides in-depth explanations, tips, and **real-world scenarios** to make the commands more useful. It's designed for both beginners (with Chapter 1 for those new to the command line) and experienced users.

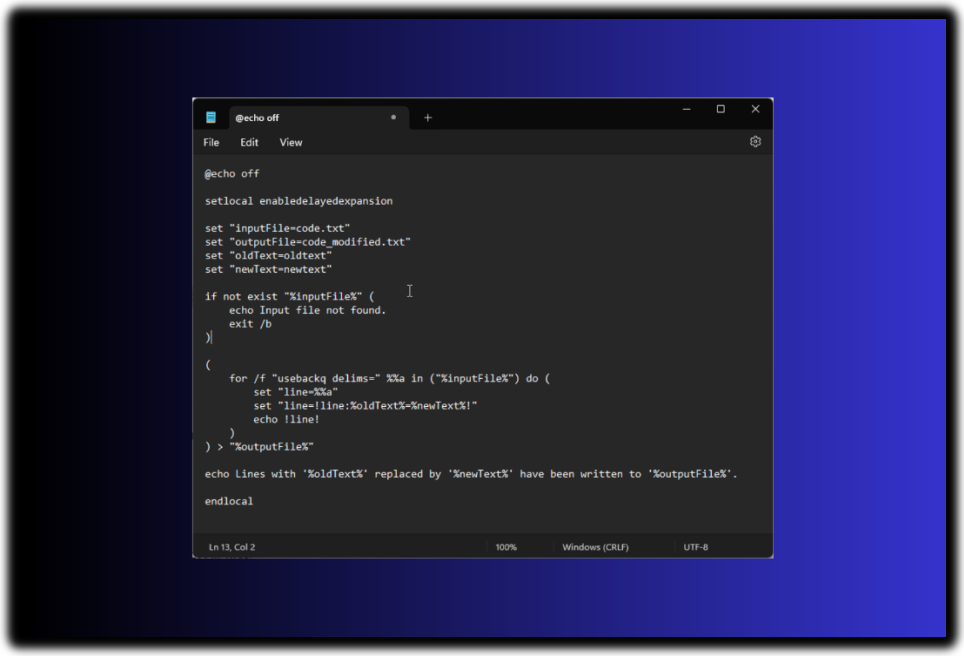


Real World Scenarios:

Each command and utility is backed by real-world stories, offering **practical examples** of how and why to use them, as well as potential pitfalls to watch out for.

Scripts and Batch Files:

The author provides scripts and batch files based on personal experience, which readers can use or adapt for their needs. These tools are presented as real-world, tested solutions.



THE BOOK AND WHAT YOU'LL LEARN

Broad Appeal (Chapters 1-5):

The initial chapters are designed for anyone who uses Windows, regardless of their technical expertise. These chapters will introduce powerful command-line tools that can simplify everyday tasks.

Basic Windows Knowledge Assumed:

While the early chapters are for everyone, the book assumes you have a foundational understanding of how to use Windows, such as navigating files and using basic input devices. It jumps directly into command-line usage without covering Windows basics.

Developer and Administrator Focus (Chapters 6-17):

These chapters delve into more advanced topics specifically for developers and system administrators. This includes commands for tasks not possible in the GUI, **accessing functions within DLL files**, automating tasks, using helpful third-party applications, and exploring the .NET Framework (like compiling JScript).

Looking Ahead to Vista and PowerShell (Chapters 18-19):

The final chapters provide a glimpse into the command-line experience in Windows Vista and **introduce Windows PowerShell**. PowerShell is presented as a significant advancement, offering more power, efficiency, and better security, similar to UNIX shells. It's also mentioned that PowerShell can be downloaded for older versions of Windows.

STANDARD WINDOWS UTILITIES

🔧 Standard Windows Utilities — Chapter Breakdown

Ch. 1: Using the Command Line Effectively
Ch. 2: Completing Data-Specific Tasks
Ch. 3: Discovering the System Status
Ch. 4: Locating Files & Other Resources
Ch. 5: Securing & Monitoring a System
Ch. 6: Developer & Low-Level Utilities

📘 Chapter 1: Using the Command Line Effectively

🧠 Why It Matters

- Command line = power tool. GUI = pretty but limited.
- The command line lets you work faster, automate, and **make fewer mistakes**.
- Many critical system features aren't in the GUI at all—only accessible from the CLI.

🗨 *"It's not outdated—it's underground."*

📉 Why People Dropped It

- GUI made computers more user-friendly—but it didn't replace everything.
- Command line seems harder (syntax, memorization), but modern CLI is more scriptable and forgiving.
- Most people don't realize how much is **missing** from the GUI or how **many tasks** are easier from the CLI.

🔄 Evolution of the Command Line

- DOS: full command line days.
- Windows: GUI took over, but the command line **never disappeared**.
- Vista: introduced **tons of new CLI tools**, despite the rise of PowerShell.

⚡ *This book covers ~280 command line utilities (standard + Vista-specific).*

💡 What You Can Do with It

- Automate tasks with **batch files or scripts** (no need to remember commands every time).
- Run commands in background, on schedule, or without user input.
- Replace 3rd-party tools (especially sketchy search tools) with native ones like **FindStr** and **Dir**.

🛠 Tools & Concepts Introduced in Chapter 1

- **Viewing commands by purpose** (instead of memorizing alphabetically)
- **Internal vs External commands** (Internal = built into cmd.exe, e.g., dir, cd; External = separate .exe files)
- **Customizing the command window** (font, size, colors, scrollback, etc.)
- **Updating utilities via Microsoft Download Center**
- **Vista-specific command differences** (new parameters, tools)

📌 Takeaways

- CLI is about **efficiency**, **power**, and **control**.
- Batch files/scripts save time and remove repetition.
- You already have a goldmine of tools on your system—this book helps you dig them out.

📌 Understanding Why the Command Line Is So Important

🖼 A Little History

- **Windows was born** to:
  - ✓ Allow multitasking (DOS was one-app-at-a-time).
  - ✓ Be more *"user-friendly"* (no need to memorize commands).
- GUI succeeded in making interactive apps easier (word processors, dev tools).
- But many **noninteractive/background tasks** are still better handled with CLI.

🛠 When GUI Just Ain't It

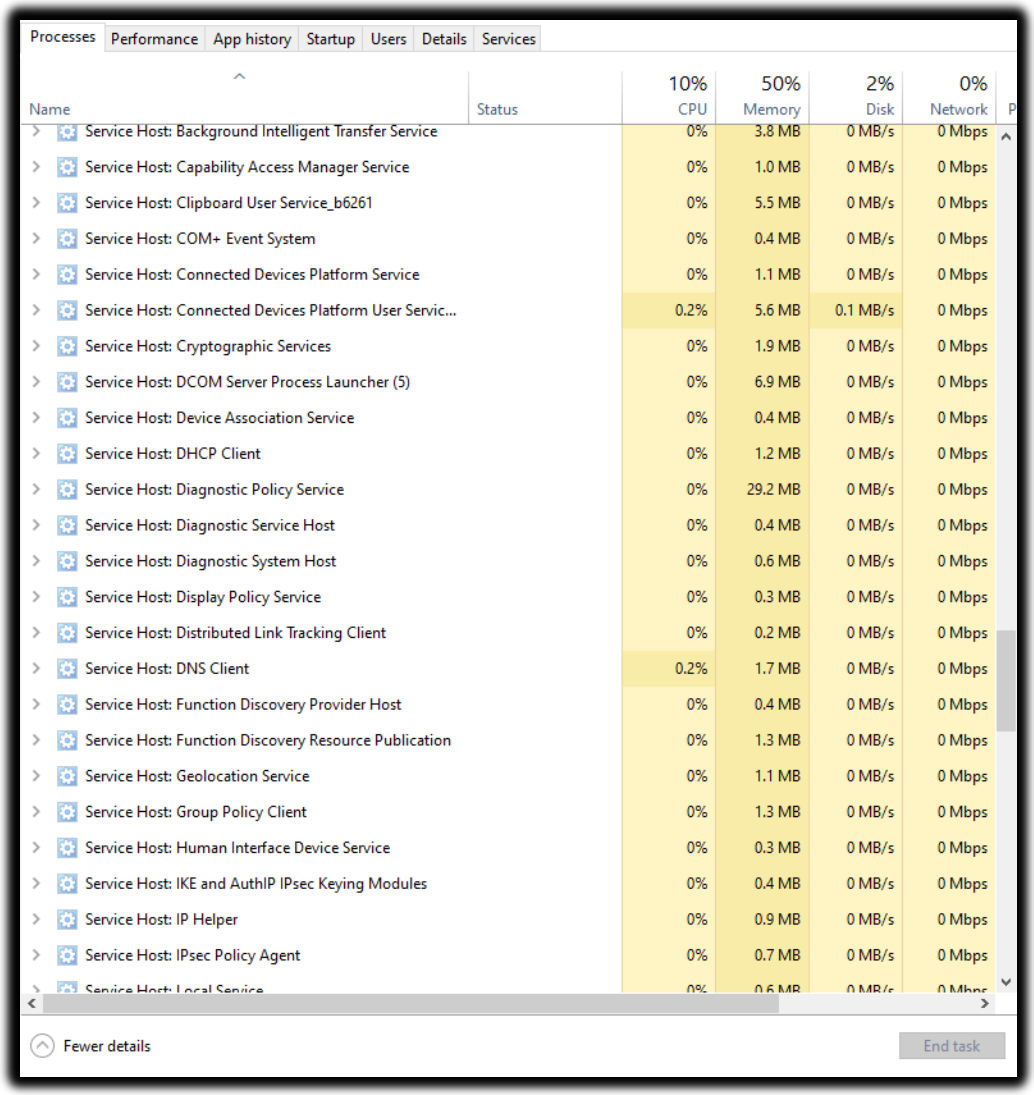
- Example: **Backups**.
  - ✓ Should run in background.
  - ✓ Need speed, accuracy, and repeatability.
  - ✓ GUI = not ideal (you're using files, it slows down, and not easily automated).
  - ✓ CLI = perfect (no need for user input, can run scheduled).
- CLI > GUI for **automation**.

🔍 Real Talk: Search is Broken in Explorer

- GUI search is **slow**, **inaccurate**, and **doesn't index** everything (even on Vista).
- Want to search inside binaries or random file types? GUI fails.
- **CLI tool like FindStr** crushes it.
  - ✓ Searches **inside any file**, including executables.
  - ✓ Fast, direct, and accurate.

🕵️ Security: What's Actually Running?

- **GUI's Task Manager:**
  - ✓ Limited view (SvcHost entries are vague).
  - ✓ Even Window's "Services" tab helps *a little*.



Processes | Performance | App history | Startup | Users | Details | Services

Name	Status	10% CPU	50% Memory	2% Disk	0% Network
Service Host: Background Intelligent Transfer Service		0%	3.8 MB	0 MB/s	0 Mbps
Service Host: Capability Access Manager Service		0%	1.0 MB	0 MB/s	0 Mbps
Service Host: Clipboard User Service_b6261		0%	5.5 MB	0 MB/s	0 Mbps
Service Host: COM+ Event System		0%	0.4 MB	0 MB/s	0 Mbps
Service Host: Connected Devices Platform Service		0%	1.1 MB	0 MB/s	0 Mbps
Service Host: Connected Devices Platform User Servic...		0.2%	5.6 MB	0.1 MB/s	0 Mbps
Service Host: Cryptographic Services		0%	1.9 MB	0 MB/s	0 Mbps
Service Host: DCOM Server Process Launcher (5)		0%	6.9 MB	0 MB/s	0 Mbps
Service Host: Device Association Service		0%	0.4 MB	0 MB/s	0 Mbps
Service Host: DHCP Client		0%	1.2 MB	0 MB/s	0 Mbps
Service Host: Diagnostic Policy Service		0%	29.2 MB	0 MB/s	0 Mbps
Service Host: Diagnostic Service Host		0%	0.4 MB	0 MB/s	0 Mbps
Service Host: Diagnostic System Host		0%	0.6 MB	0 MB/s	0 Mbps
Service Host: Display Policy Service		0%	0.3 MB	0 MB/s	0 Mbps
Service Host: Distributed Link Tracking Client		0%	0.2 MB	0 MB/s	0 Mbps
Service Host: DNS Client		0.2%	1.7 MB	0 MB/s	0 Mbps
Service Host: Function Discovery Provider Host		0%	0.4 MB	0 MB/s	0 Mbps
Service Host: Function Discovery Resource Publication		0%	1.3 MB	0 MB/s	0 Mbps
Service Host: Geolocation Service		0%	1.1 MB	0 MB/s	0 Mbps
Service Host: Group Policy Client		0%	1.3 MB	0 MB/s	0 Mbps
Service Host: Human Interface Device Service		0%	0.3 MB	0 MB/s	0 Mbps
Service Host: IKE and AuthIP IPsec Keying Modules		0%	0.4 MB	0 MB/s	0 Mbps
Service Host: IP Helper		0%	0.9 MB	0 MB/s	0 Mbps
Service Host: IPsec Policy Agent		0%	0.7 MB	0 MB/s	0 Mbps
Service Host: Local Service		0%	0.6 MB	0 MB/s	0 Mbps

Fewer details | End task

- **CLI's TaskList:**
  - ✓ Shows exact services running **inside** each SvcHost.
  - ✓ Reveals **PID** (Process ID) so you can trace what's really happening.

```
Windows PowerShell
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Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Users\Nick The Chief> tasklist /svc

Image Name                     PID Services
========================= ======== ============================================
System Idle Process              0 N/A
System                           4 N/A
Registry                       108 N/A
smss.exe                       480 N/A
csrss.exe                      632 N/A
wininit.exe                    804 N/A
csrss.exe                      812 N/A
services.exe                   876 N/A
lsass.exe                      888 EFS, KeyIso, SamSs, VaultSvc
winlogon.exe                   984 N/A
svchost.exe                    688 BrokerInfrastructure, DcomLaunch, PlugPlay,
                                   Power, SystemEventsBroker
fontdrvhost.exe                816 N/A
fontdrvhost.exe                504 N/A
WUDFHost.exe                  1084 N/A
svchost.exe                   1164 RpcEptMapper, RpcSs
svchost.exe                   1212 LSM
svchost.exe                   1304 lmhosts
svchost.exe                   1364 NcbService
svchost.exe                   1376 TimeBrokerSvc
svchost.exe                   1408 EventLog
svchost.exe                   1488 hidserv
dwm.exe                       1528 N/A
svchost.exe                   1616 nsi
svchost.exe                   1672 Dhcp
svchost.exe                   1696 SEMgrSvc
svchost.exe                   1816 Schedule
svchost.exe                   1880 ProfSvc
svchost.exe                   1940 NlaSvc
svchost.exe                   2044 UserManager
atiesrxx.exe                  1148 AMD External Events Utility
svchost.exe                   1748 camsvc
svchost.exe                   2084 CoreMessagingRegistrar
svchost.exe                   2092 netprofm
helperservice.exe             2200 N/A
svchost.exe                   2224 StateRepository
svchost.exe                   2344 DispBrokerDesktopSvc
conhost.exe                   2432 N/A
svchost.exe                   2440 WinHttpAutoProxySvc
svchost.exe                   2536 Dnscache
atieclxx.exe                  2632 N/A
```

🧠 You can't defend your system if you don't know what it's doing. CLI tells the truth, GUI hides it.

🧠 CLI: Your X-Ray Goggles for Windows

🌐 Network & Web Intelligence

- **DNS Who?** → Use CLI to resolve exact IP addresses behind friendly URLs.
  - ✓ Example: nslookup openai.com = no more guessing who's spoofing your emails.
- **Slow Internet?** → Diagnose if it's your Wi-Fi, your ISP, or a bad route on the web.
  - ✓ Tools like ping, tracert, and pathping tell the truth before you waste an hour on customer support.

```
PS C:\Users\Nick The Chief> nslookup telegram.org
Server:  UnKnown
Address:  192.168.110.1

Non-authoritative answer:
Name:    telegram.org
Addresses:  2001:67c:4e8:f004::9
          149.154.167.99

PS C:\Users\Nick The Chief> nslookup nick.com
Server:  UnKnown
Address:  192.168.110.1

Non-authoritative answer:
Name:    nick.com
Addresses:  2600:1f18:297:ba24:b508:76de:f4a:df34
          2600:1f18:297:ba0e:7900:f622:48a8:f266
```

🏠 Local + Network Resource Discovery

- CLI doesn't just talk to the web—it **maps your local network** too.
  - ✓ netstat, ipconfig, nbtstat, arp, and friends help you find rogue devices, hidden services, and weird traffic.
  - ✓ Forget the GUI's "Network" tab—it only shows happy paths. CLI shows **everything**, including what *shouldn't* be there.

💻 Deep System Visibility

- The command line sees stuff the GUI pretends doesn't exist. You want actual disk usage? Real-time CPU info? Driver status? It's all there.
- CLI **doesn't assume** what you're trying to do—it just **does what you tell it**.

```
Windows PowerShell
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Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Users\Nick The Chief> fsutil volume diskfree C:
Total free bytes        : 109,845,753,856 (102.3 GB)
Total bytes             : 179,373,850,624 (167.1 GB)
Total quota free bytes  : 109,845,753,856 (102.3 GB)
PS C:\Users\Nick The Chief> wmic cpu get LoadPercentage
LoadPercentage
6

PS C:\Users\Nick The Chief> driverquery /v

Module Name  Display Name           Description            Driver Type   Start Mode State      Status     Accept Stop Accept Pause Paged Pool(bytes)
            Path                                              Init(bytes)
=========== ===================== ===================== ============ ========== ========== ========== =========== =========== ================
=========== ============================================ ===========
1394ohci     1394 OHCI Compliant Ho 1394 OHCI Compliant Ho Kernel        Manual     Stopped    OK         FALSE       FALSE       4,096
            C:\Windows\system32\drivers\1394ohci.sys         4,096
3ware        3ware                  3ware                 Kernel        Manual     Stopped    OK         FALSE       FALSE       0
:28:03 AM   C:\Windows\system32\drivers\3ware.sys            4,096
ACPI         Microsoft ACPI Driver  Microsoft ACPI Driver  Kernel        Boot       Running    OK         TRUE        FALSE       176,128
            C:\Windows\system32\drivers\ACPI.sys             24,576
AcpiDev      ACPI Devices driver    ACPI Devices driver    Kernel        Manual     Stopped    OK         FALSE       FALSE       8,192
            C:\Windows\system32\drivers\AcpiDev.sys          4,096
acpiex       Microsoft ACPIEx Drive Microsoft ACPIEx Drive Kernel        Boot       Running    OK         TRUE        FALSE       40,960
            C:\Windows\system32\Drivers\acpiex.sys           4,096
acpipagr     ACPI Processor Aggrega ACPI Processor Aggrega Kernel        Manual     Stopped    OK         FALSE       FALSE       4,096
            C:\Windows\system32\drivers\acpipagr.sys         4,096
```

🏴‍☠️ Why CLI Is Essential (Even If You Don't Use It Daily)

GUI	CLI
Babysits you with buttons	Treats you like an admin
Shows what it thinks you *need*	Shows **everything**, even the risky stuff
Slows down with animations	Lightning-fast execution, no graphics to slow it down
Hides complex routes & internals	Exposes true paths, ports, and performance

⚠️ Reality check:

If you're serious about system security or performance, you **can't trust** the GUI alone. *The CLI ain't dead. It's just quiet.* And like most quiet things in tech, it's where the **power** lives. You don't *have* to use it every day... but you should know it like a sidearm.

🎯 Command Line = NOT That Deep (If You Know the Vibe)

🤯 What People Think the CLI Is:

"You just type a bunch of cryptic stuff, hope you don't brick your machine, then stare at white text on a black screen like a hacker in a bad movie."

Wrong. That's the **myth**.

🚪 CLI Is Easy If You Know the Secret Knock

You don't need to memorize commands like it's the SAT.
You need **just one trick**:

🔑 **Add /? to *any* command to get instant help.**

🤯 **That's it.**

- **tasklist /?**
- **xcopy /?**
- **netstat /?**
- **shutdown /?**

Almost *every* built-in Windows CLI tool responds to that magic switch and dumps a cheat sheet right in your terminal.

You'll see all the **options** the command supports (scrollable if it's long).

🧠 Why You Should Use Commands Directly First

Yeah, scripting and batch files are cool...
But before you automate anything, you've gotta understand:

- What each command does
- What input it needs
- What output it gives

Once you *get* a few commands, stringing them together into automation is ✨child's play✨.

⚡ Real Talk

If you know how to call a command with /? and read its output, you're already ahead of 90% of Windows users. Wanna keep the momentum going? Next up is **how to view commands by purpose**—so instead of guessing what tools exist, you'll know where each one fits in the big picture (file ops, networking, diagnostics, etc.).

```
PS C:\Users\Nick The Chief> tasklist /?

TASKLIST [/S system [/U username [/P [password]]]]
         [/M [module] | /SVC | /V] [/FI filter] [/FO format] [/NH]

Description:
    This tool displays a list of currently running processes on
    either a local or remote machine.

Parameter List:
   /S     system           Specifies the remote system to connect to.

   /U     [domain\]user    Specifies the user context under which
                           the command should execute.

   /P     [password]       Specifies the password for the given
                           user context. Prompts for input if omitted.

   /M     [module]         Lists all tasks currently using the given
                           exe/dll name. If the module name is not
                           specified all loaded modules are displayed.

   /SVC                    Displays services hosted in each process.

   /APPS                   Displays Store Apps and their associated processes.
   /V                      Displays verbose task information.

   /FI    filter           Displays a set of tasks that match a
                           given criteria specified by the filter.

   /FO    format           Specifies the output format.
                           Valid values: "TABLE", "LIST", "CSV".

   /NH                     Specifies that the "Column Header" should
                           not be displayed in the output.
                           Valid only for "TABLE" and "CSV" formats.

   /?                      Displays this help message.
```

```
PS C:\Users\Nick The Chief> netstat /?

Displays protocol statistics and current TCP/IP network connections.

NETSTAT [-a] [-b] [-e] [-f] [-n] [-o] [-p proto] [-r] [-s] [-t] [-x] [-y] [interval]

  -a            Displays all connections and listening ports.
  -b            Displays the executable involved in creating each connection or
                listening port. In some cases well-known executables host
                multiple independent components, and in these cases the
                sequence of components involved in creating the connection
                or listening port is displayed. In this case the executable
                name is in [] at the bottom, on top is the component it called,
                and so forth until TCP/IP was reached. Note that this option
                can be time-consuming and will fail unless you have sufficient
                permissions.
  -e            Displays Ethernet statistics. This may be combined with the -s
                option.
  -f            Displays Fully Qualified Domain Names (FQDN) for foreign
                addresses.
  -n            Displays addresses and port numbers in numerical form.
  -o            Displays the owning process ID associated with each connection.
  -p proto      Shows connections for the protocol specified by proto; proto
                may be any of: TCP, UDP, TCPv6, or UDPv6.  If used with the -s
                option to display per-protocol statistics, proto may be any of:
                IP, IPv6, ICMP, ICMPv6, TCP, TCPv6, UDP, or UDPv6.
  -q            Displays all connections, listening ports, and bound
                nonlistening TCP ports. Bound nonlistening ports may or may not
                be associated with an active connection.
  -r            Displays the routing table.
  -s            Displays per-protocol statistics.  By default, statistics are
                shown for IP, IPv6, ICMP, ICMPv6, TCP, TCPv6, UDP, and UDPv6;
                the -p option may be used to specify a subset of the default.
  -t            Displays the current connection offload state.
  -x            Displays NetworkDirect connections, listeners, and shared
                endpoints.
  -y            Displays the TCP connection template for all connections.
                Cannot be combined with the other options.
  interval      Redisplays selected statistics, pausing interval seconds
                between each display.  Press CTRL+C to stop redisplaying
                statistics.  If omitted, netstat will print the current
                configuration information once.
```

COMMANDLINE SWITCHES AND PIPES

Think of a command-line switch as an **option** or a **modifier** that you add to a command to change how it behaves or what information it displays. It's like adding extra instructions to the main command.

Switches allow you to customize the command's output or its actions. They let you be more specific about what you want the command to do.

Syntax: As the book mentioned, switches usually start with a forward slash (/) or sometimes a dash (-) followed by a letter or a word.

Dir

If you just type **dir** and press Enter, it will show a basic list. However, if you want to see more details, like the file size and the last modified date, you can use the **/W switch** (for "wide" format) or the **/A switch** (to display files with specific attributes). So, Dir /W would show a wider listing, and Dir /A:H would show hidden files.

```
C:\Users\Nick The Chief\Downloads>dir /a:h
 Volume in drive C has no label.
 Volume Serial Number is BCC5-4691

 Directory of C:\Users\Nick The Chief\Downloads

04/09/2025  04:19 PM               282 desktop.ini
04/11/2025  07:31 PM               162 ~$eManOfSteel2 - NixonSchool92.docx
04/11/2025  03:12 PM               162 ~$eWorld.docx
               3 File(s)            606 bytes
               0 Dir(s)  109,825,507,328 bytes free

C:\Users\Nick The Chief\Downloads>dir /a:s
 Volume in drive C has no label.
 Volume Serial Number is BCC5-4691

 Directory of C:\Users\Nick The Chief\Downloads

04/09/2025  04:19 PM               282 desktop.ini
               1 File(s)            282 bytes
               0 Dir(s)  109,825,445,888 bytes free

C:\Users\Nick The Chief\Downloads>dir /a
 Volume in drive C has no label.
 Volume Serial Number is BCC5-4691

 Directory of C:\Users\Nick The Chief\Downloads

04/12/2025  07:49 AM    <DIR>          .
04/12/2025  07:49 AM    <DIR>          ..
11/11/2024  07:44 AM        31,133,896 1 ... Operating Systems.pdf
11/02/2024  11:51 AM        13,946,903 2 ... TCP-IP-Vol1.pdf
11/02/2024  11:52 AM        39,820,078 3 ... TCP-IP-Vol2.pdf
11/30/2024  04:59 PM        16,110,511 4 ... CommandLine Administration Best2.pdf
```

Pipes (|):

A **pipe is a symbol (|)** used on the command line to **send the output of one command as the input to another command.** It's like creating a connection between two commands, where the first command's result becomes what the second command works with.

Let's say you want to see a list of all the files in a directory, but you only want to see the ones that contain the word "Gemini" in their name.

```
C:\Users\Nick The Chief\Downloads>dir | findstr "Gemini"
04/11/2025  11:44 PM           539,008 Gemini_Generated_Image_idzou1idzou1idzo.jpg
04/11/2025  11:45 PM           630,299 Gemini_Generated_Image_lffprlffprlffprl.jpg
```

Switches modify the behavior of a single command. **Use / or -**

Pipes connect the output of one command to the input of another, allowing you to chain commands together for more complex operations. **Use |**

⚡ COMMAND LINE SYNTAX

Most Windows CLI commands follow this format:

```
Command [options] [arguments]
```

Understanding Optional Parameters (Square Brackets [ ])

In command syntax, **square brackets** [ ] mean that the enclosed parameter is **optional**—you can include it or omit it.

Example:

```
C:\Users\Nick The Chief\Downloads>tasklist [/S system [/U username [/P [password]]]]
```

Means:

You can run just tasklist with or without options. Some might might misinterpret it as you write the command with the brackets, which is wrong, there's nothing like writing the commands with brackets included. It means:

- Or add **/S** system (optional).
- If you use /S, you can optionally add **/U** username.
- If you use /U, you can optionally add **/P** password.

The square brackets ([ ]) are **just documentation syntax** — not something you actually write. They mean, *"This part is optional."* They **do not** mean "copy me into your command."

✅ Correct Ways to Run It:

1. **Basic Usage (no options)**

cmd Copy
```
tasklist
```

- Lists all running processes on your current machine.

2. **With a Remote System (`/S`)**

cmd Copy
```
tasklist /S 192.168.1.100
```

- Lists processes on a remote computer (IP/hostname).

3. **With Username (`/U`)**

cmd Copy
```
tasklist /S 192.168.1.100 /U Admin
```

- Connects to `192.168.1.100` using the username `Admin`.

4. **With Password (`/P`)**

cmd Copy
```
tasklist /S 192.168.1.100 /U Admin /P MyPassword123
```

- Provides the password for authentication.

5. **Password Prompt (if `/P` is empty)**

cmd Copy

```
tasklist /S 192.168.1.100 /U Admin /P
```

- You'll be **prompted** to enter the password (more secure).

You can do this with Virtual Machines which is a simpler way to explore. Let's do it, I have some VMs ready.

❌ **Incorrect Ways (What NOT to Do)**

- **Including `[ ]` in the command:**

cmd Copy

```
tasklist [/S 192.168.1.100]  ❌ (Wrong!)
```

  - The brackets are just notation—don't type them!

- **Skipping required parts of a switch:**

cmd Copy

```
tasklist /U Admin  ❌ (Missing /S system)
```

  - If you use `/U`, you **must** have `/S` first.

1. *[ ] means Optional switch(include or exclude as needed).*
2. *Never type [ ] in the actual command.*
3. *Order matters!*
   - */S must come before /U.*
   - */U must come before /P.*
4. *If /P has no password, you'll be prompted.*
5. *Windows prefers /, not - like Linux.*
6. *But PowerShell often accepts both.*

PARAMETERS

Parameters are just values you provide to a command ***without* using a / or – flag** like in switches. They tell the command *what* to act upon.

Certain ***command-line commands accept a drive letter*** (followed by a colon, like D:) as a parameter to specify which storage location the command should operate on.

```
C:\Users\Nick The Chief\Downloads>dir D:
 Volume in drive D is ToshibaX
 Volume Serial Number is F8D9-AC95

 Directory of D:\

11/11/2024  07:44 AM        31,133,896 1 ... Operating Systems.pdf
11/02/2024  11:51 AM        13,946,903 2 ... TCP-IP-Vol1.pdf
11/02/2024  11:52 AM        39,820,078 3 ... TCP-IP-Vol2.pdf
03/02/2025  02:30 AM        17,839,384 5 ... Automate the Boring Stuff with Python.pdf
01/31/2025  03:10 PM       294,288,017 9 Uncommon Ways to GET and STAY Rich (Watch and Share This many times).mp4
04/08/2025  02:33 PM         1,166,138 A-Farewell-to-Virology-Expert-Edition.pdf
04/11/2025  06:53 PM    <DIR>          Action!
05/15/2024  01:35 AM       518,817,795 Best of Emachichi,3 Hours Nonstop Mix, Speech Entertainment.mp4
03/19/2025  05:08 PM       228,434,309 calm your heart..mp4
04/09/2025  03:47 AM           926,454 DevonIsFreed.png
04/04/2025  12:32 AM     2,947,857,762 Dune.2021.1080p.WEBRip.x264.AAC5.1-[YTS.MX].mp4
04/04/2025  12:37 AM     2,968,283,738 Dune.Part.Two.2024.1080p.WEBRip.x264.AAC-[YTS.MX].mp4
03/25/2025  04:47 AM     2,722,103,296 EternalNotes0.hc
03/31/2025  01:30 AM     3,246,391,296 EternalNotes1.hc
03/31/2025  07:56 PM     1,073,741,824 EternalNotes2.hc
03/31/2025  07:56 PM     1,073,741,824 EternalNotes3.hc
03/31/2025  11:20 PM     2,357,198,848 EternalNotes5.hc
04/11/2025  06:53 PM    <DIR>          EverybodyHatesChris
04/01/2025  06:05 PM     6,442,450,944 FamilyPhotos.hc
04/11/2025  03:49 AM    <DIR>          GeminiAI.Images
04/09/2025  03:14 PM    <DIR>          Github.New
```

Here, **D:** is the parameter telling the **cd** command *where* you want to go.

If you want to see the files and folders at the root of your D: drive, you can use:

```
C:\Users\Nick The Chief\Downloads>D:

D:\>dir | findstr ".hc"
03/25/2025  04:47 AM     2,722,103,296 EternalNotes0.hc
03/31/2025  01:30 AM     3,246,391,296 EternalNotes1.hc
03/31/2025  07:56 PM     1,073,741,824 EternalNotes2.hc
03/31/2025  07:56 PM     1,073,741,824 EternalNotes3.hc
03/31/2025  11:20 PM     2,357,198,848 EternalNotes5.hc
04/01/2025  06:05 PM     6,442,450,944 FamilyPhotos.hc
04/11/2025  12:35 AM    15,305,015,296 HT.hc
01/31/2025  06:03 PM       209,715,200 Password.hc
03/31/2025  07:48 PM       364,904,448 The.727.System.hc

D:\>
```

✅ Good Starter Commands to Try `/?" On

Command	Use Case
`tasklist /?`	See active processes and services
`dir /?`	File and folder listings
`netstat /?`	Network ports and connections
`shutdown /?`	Restart, logoff, shutdown, timed power off
`robocopy /?`	Advanced file copy (better than `xcopy`)

Back to parameters (Values we pass to the actual commands).

1. Path Parameters

```
dir C:\Users\Rennex\Documents
```

No / is used—just the raw path.

2. Filename/Extension Parameter

```
dir *.txt       (Lists all .txt files)
dir report.doc  (Looks for "report.doc")
```

***.txt** and **report.doc** are **parameters** (filtering files).

3. Drive Letter Parameter

```
dir D:\        (Lists files on the D: drive)
```

D:\ is a **parameter** (the drive to scan).

📜 Key Differences Between Parameters and Switches

Feature	Parameter	Switch
Syntax	No prefix (e.g., `C:\`, `*.exe`)	Prefixed with `/` (e.g., `/W`, `/A:H`)
Purpose	Tells the command *what* to act on	Tells the command *how* to act
Usage Example	`dir C:\Photos` (target folder)	`dir /S` (include subdirectories)
Acts As	Input (location, file type, etc.)	Modifier (affects behavior/output)
Required?	Usually yes (you need to point somewhere)	Often optional (enhances control)

🧠 Quick Breakdown: Parameters vs. Switches in CMD

🔥 Parameters

- What they are: The *where* or *what* the command is working on.
- Purpose: They give the command the ***actual value*** or ***target*** it should operate on.
- Think of them as: The destination you're pointing to.
- Example:

```
dir D:\Downloads
```

➤ *"Show me what's inside the Downloads folder."*

🧭 **Parameter = destination.**

🔥 Switches

- **What they are:** The *how* the command should behave.
- **Purpose:** They ***modify*** *or* ***enhance*** what the command does.
- **Think of them as:** The **tools** you bring with you to the destination.
- **Example:**

```
dir /s
```

➤ *"Include subfolders when listing files."*

🛠️ **Switch = tools you bring with you.**

✅ Put Together:

```
dir D:\Downloads /S
```

➤ *"List everything in Downloads, including all its subfolders."*

WHY USE BATCH FILES? 📂

Batch files aren't just some relic from the MS-DOS era—they're **still super useful** for people who want to automate repetitive tasks without writing full-blown scripts or programs.

The author's point is pretty clear: **remembering every command and its switches is annoying.**

Batch files act like your **personal memory vault**—they store complex or frequently-used command sequences so you don't have to.

You write a command **once**, store it in a .bat file, and from then on? You just double-click or run the file from the command line. Done. ✅

✍️ What Is a Batch File, technically?

- A **batch file** is just a plain text file with the extension **.bat** or **.cmd.**
- It contains a list of command line commands that execute **in order**.
- You can open and edit them in any text editor (e.g., Notepad, Notepad++).

📌 Real-World Example

Let's say you have to check the running processes on a remote computer every morning.

Typing this out every time:

```
tasklist /S 192.168.1.10 /U Admin /P *
```